Tornado cash

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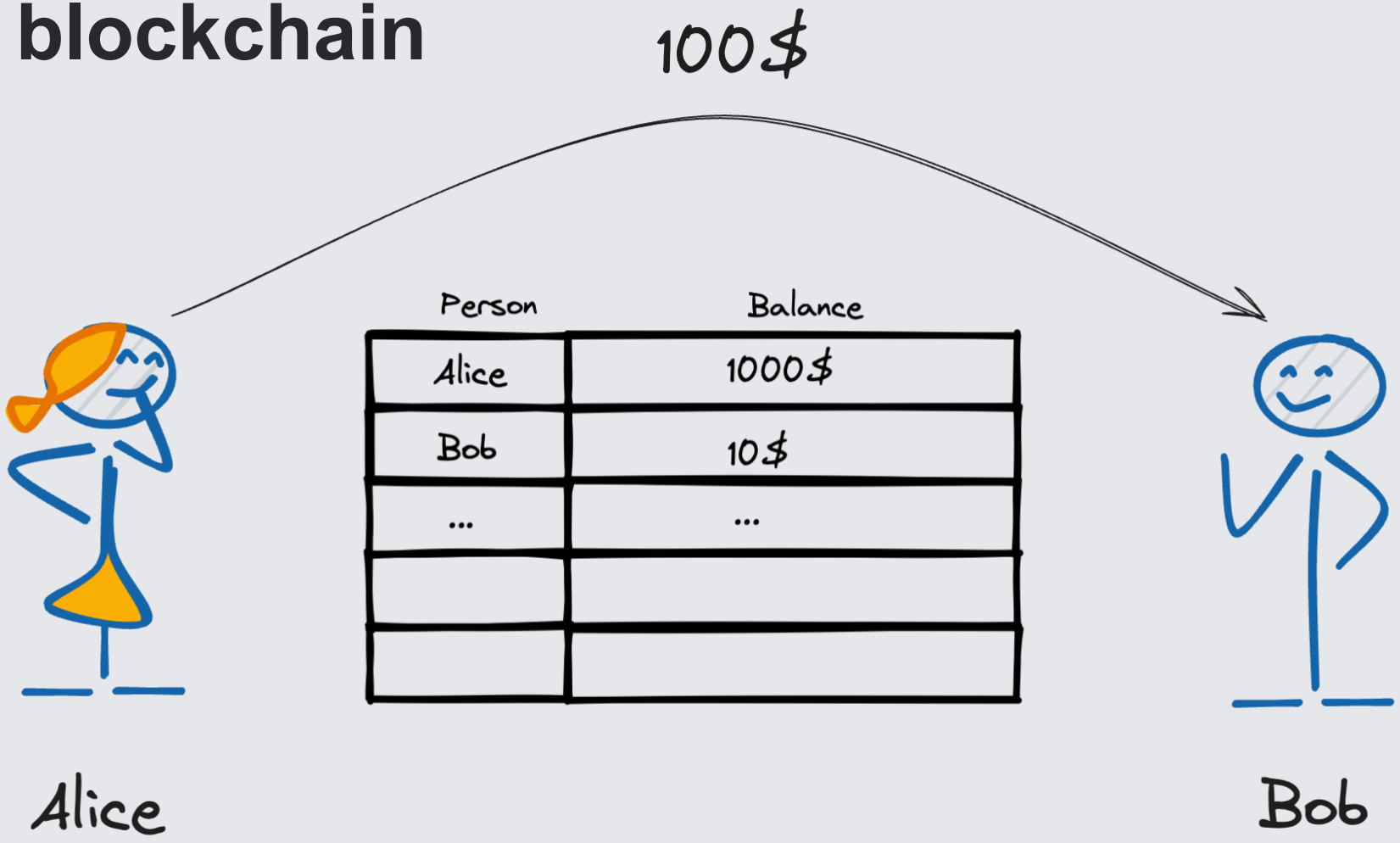
Agenda

- What is blockchain and smart contracts
- How Tornado Cash works

What is blockchain



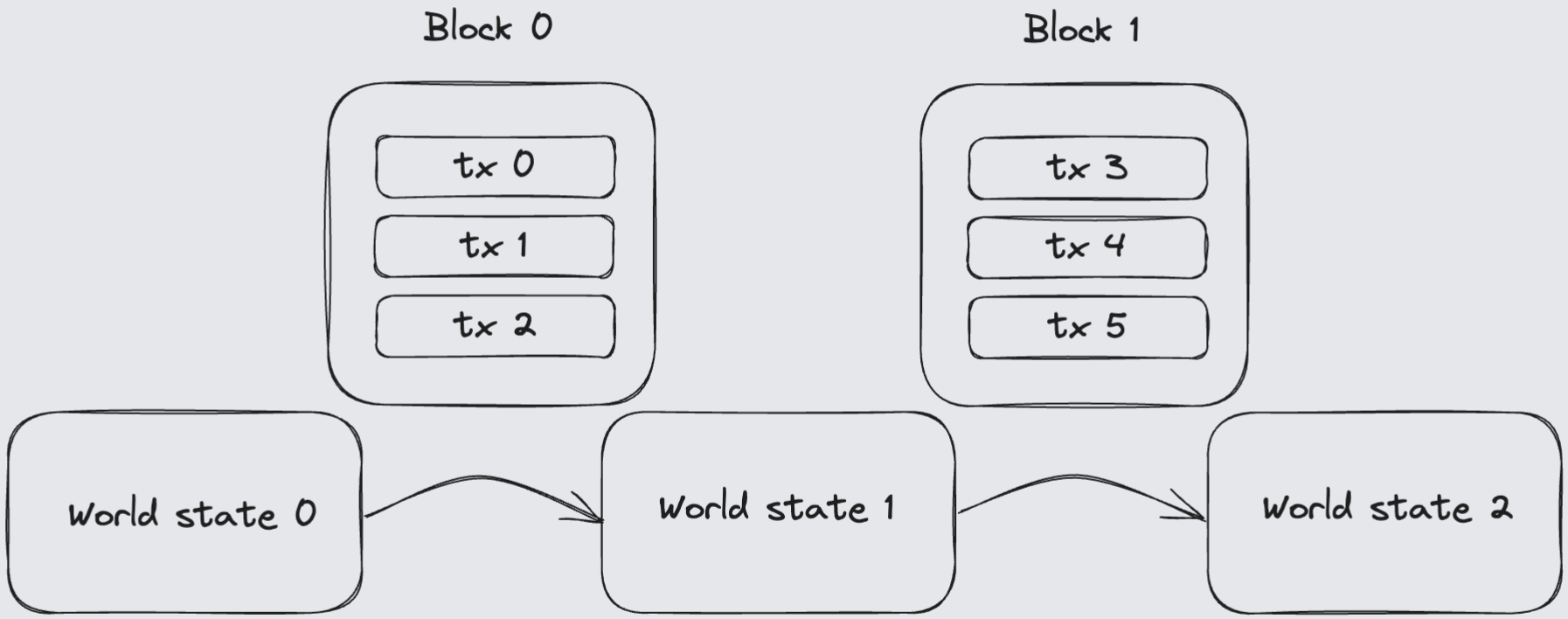
What is blockchain

Alice

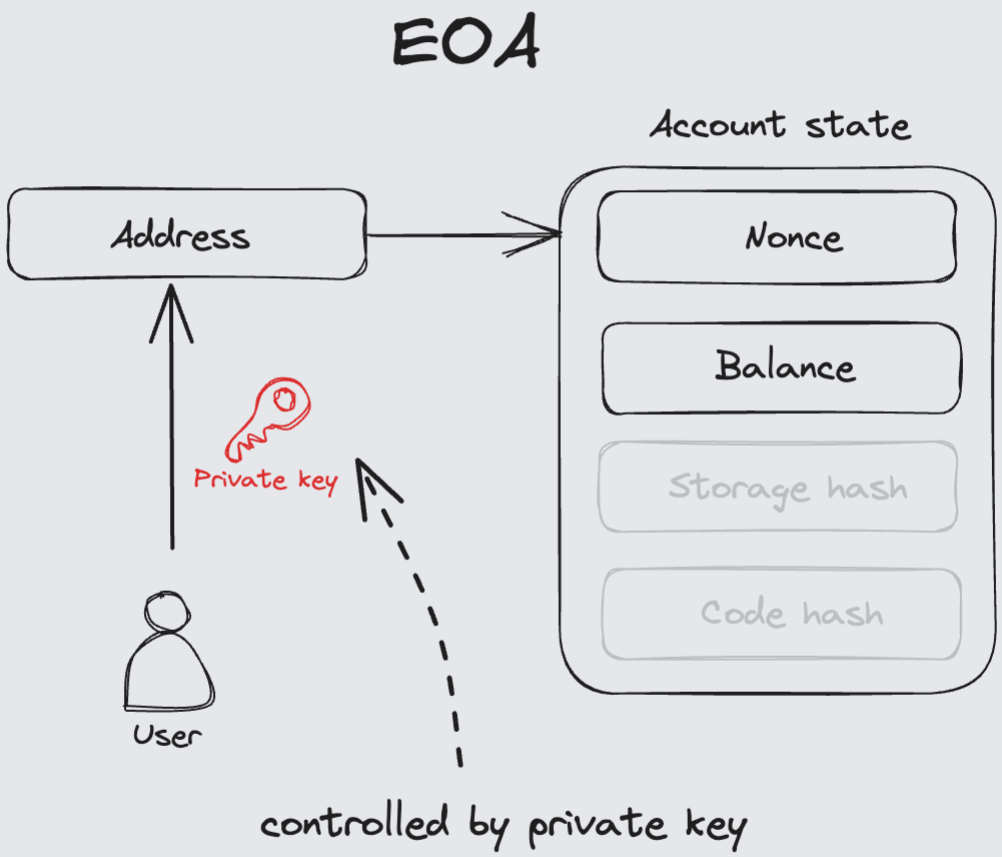
Person	Balance
Alice	900$
Bob	110$
...	...

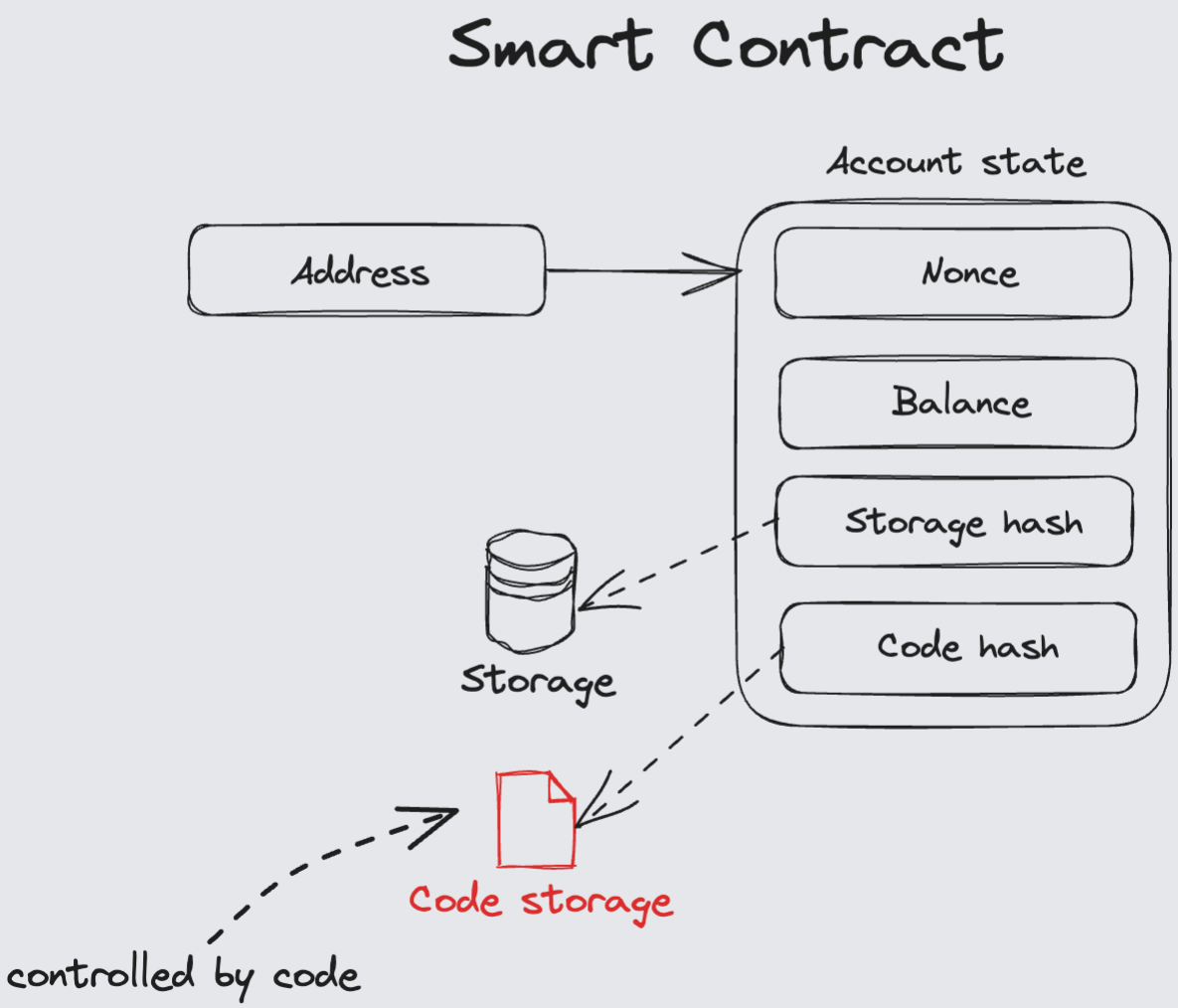
Bob

What is blockchain

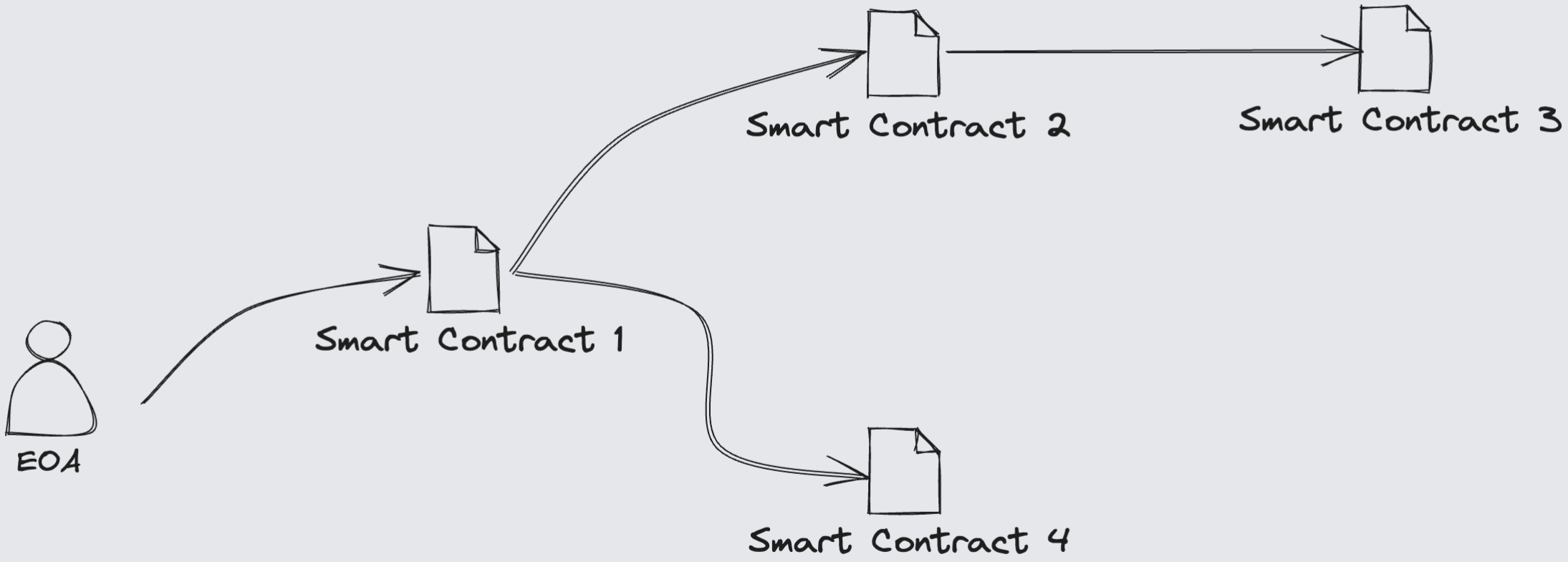


What is smart contracts

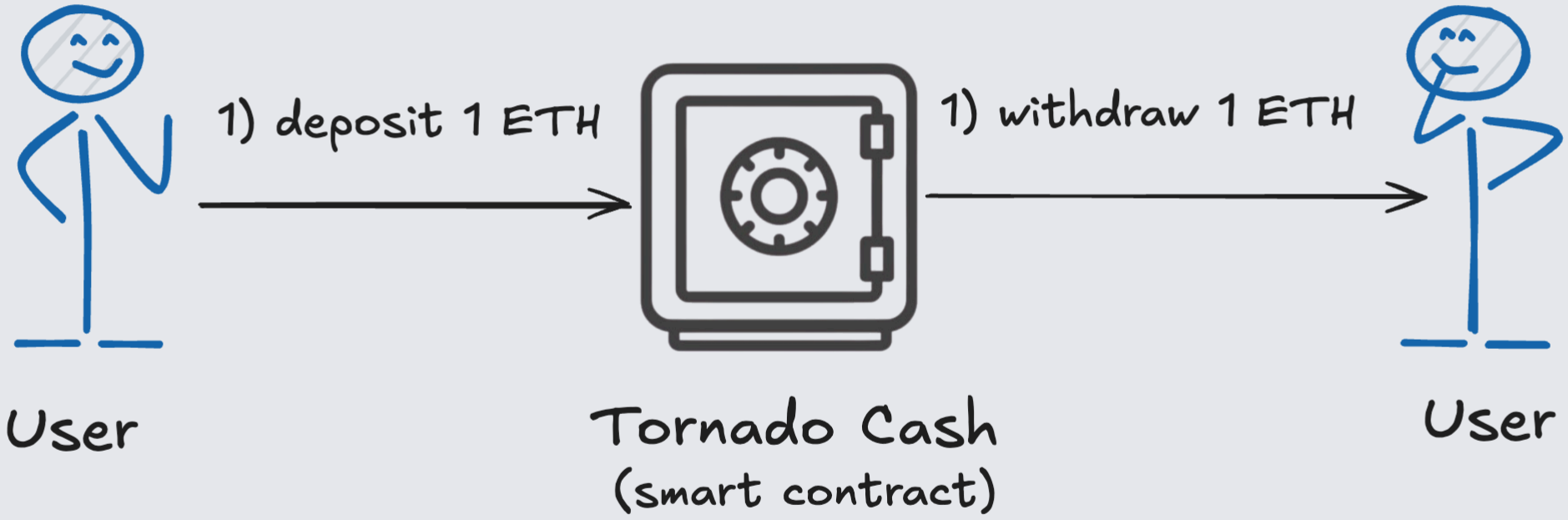




What is smart contracts

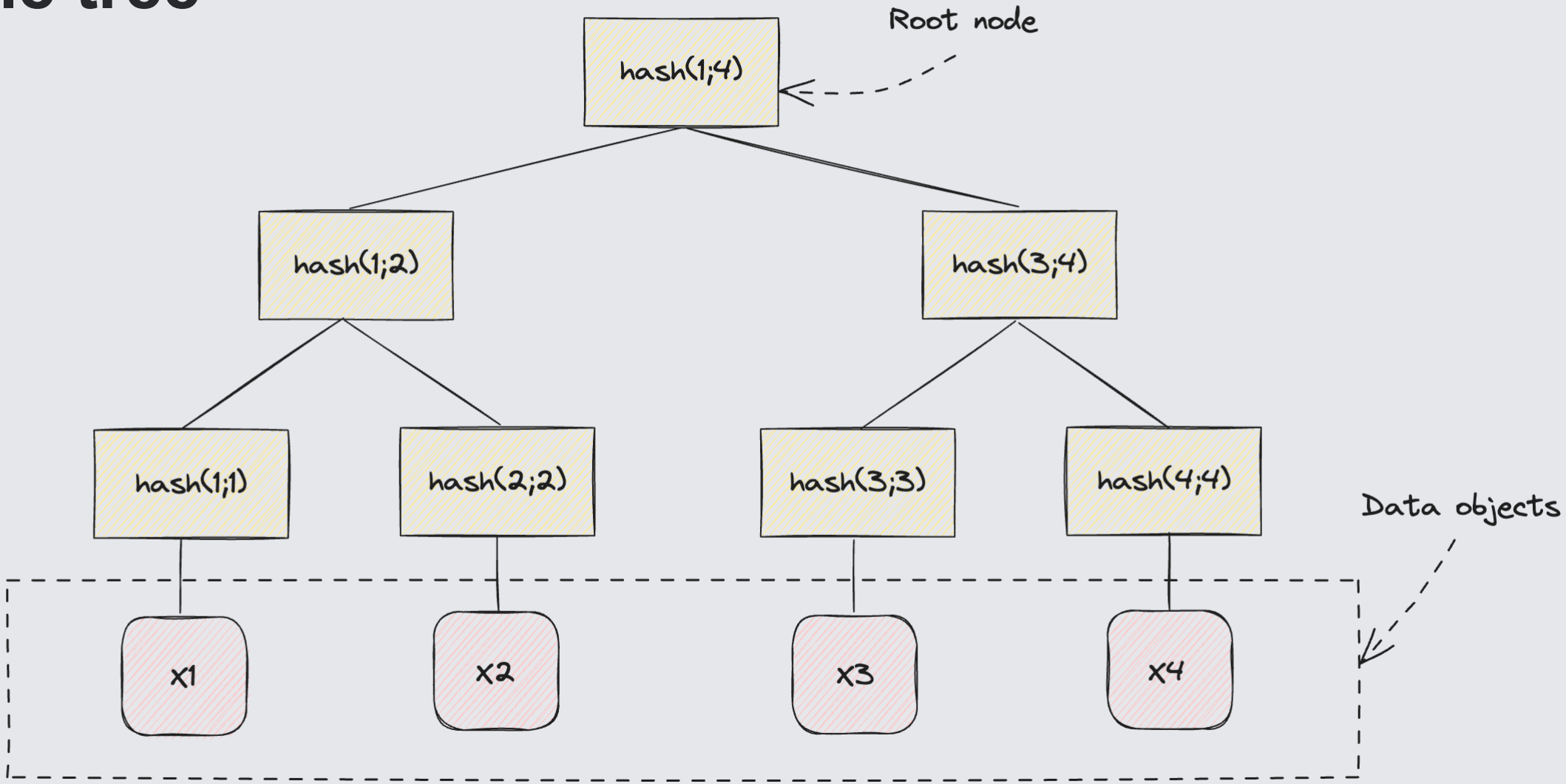


Tornado Cash



Simple implementation of Tornado Cash (not ZK)

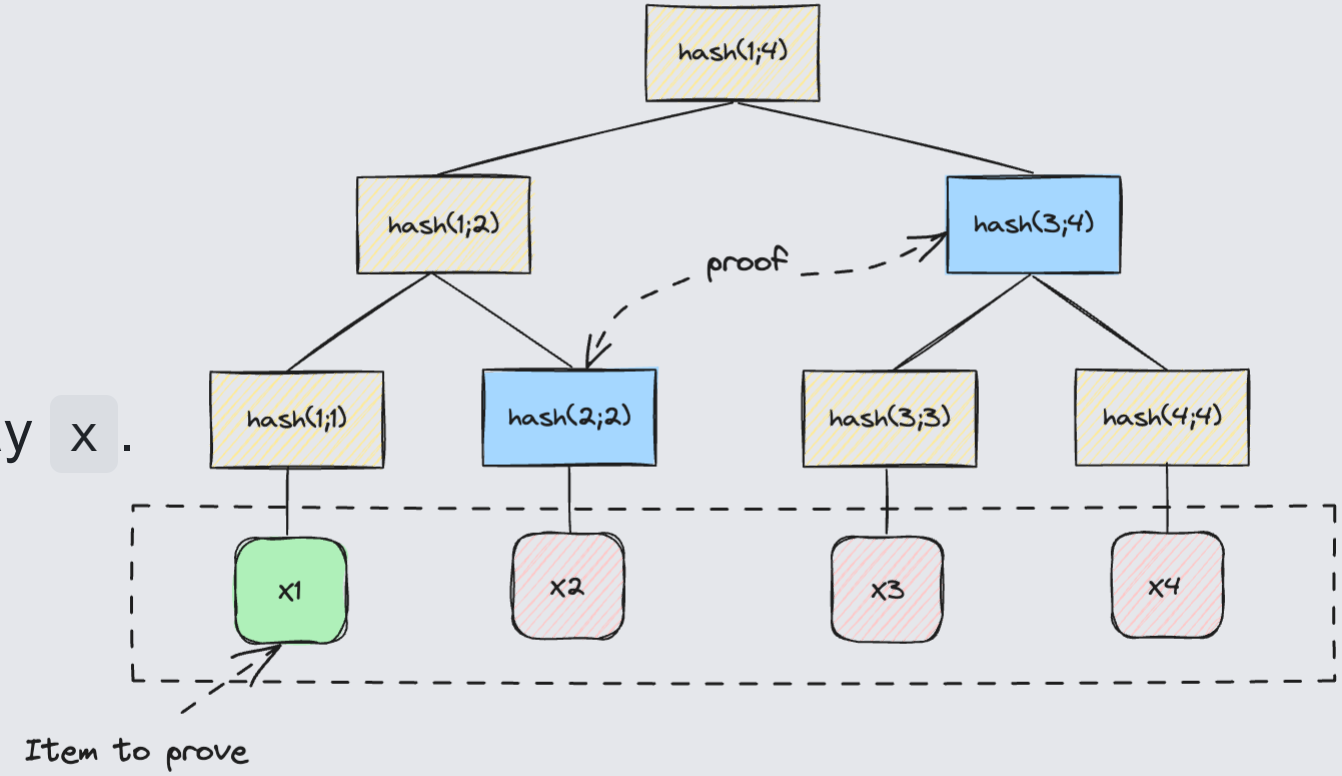
Merkle tree


Root node
hash(1;4)
hash(1;2)
hash(3;4)
hash(1;1)
hash(2;2)
hash(3;3)
hash(4;4)
Data objects
x1
x2
x3
x4

Merkle tree

Goal

- get a short representation to the array `x`.
- later prove that element i is `x[i]`.



Implementation with merkle tree (not ZK)

True implementation

Private note construction

- User randomly generates (both secrets):
  - `secret` (248-bit)
  - `nullifier` (248-bit)
- User then computes:
  - `commitment = Pedersen(nullifier || secret)`
  - `nullifierHash = Pedersen(nullifier)`
- Usage:
  - `commitment` – used in deposits, stored as leafs in Merkle tree
  - `nullifierHash` – used in withdrawals to prevent double spending